

Samanwaya Datta

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Summary

Computer Science Engineering student (2027 Batch, VIT Chennai) with hands-on experience designing scalable backend systems, RESTful APIs, and developer tooling in Go. Strong foundation in data structures, algorithms, system design, and object-oriented programming. Proven ability to deliver reliable, production-quality software in Agile team environments.

Education

Vellore Institute of Technology, Chennai

Aug 2023 – Present

B.Tech in Computer Science Engineering (Electrical & CS) — CGPA: 8.86 / 10.0 (88.6%)

St. Patrick's Higher Secondary School, Asansol

2010 – 2023

Class X: 95% — Class XII: 93%

Technical Skills

- **Languages:** Go, Python, JavaScript, C++, C, Java
- **Backend & APIs:** Node.js, Express.js, RESTful API Design, JWT Authentication, Concurrency (Go goroutines), JSON/XML Data Handling
- **AI/ML:** RAG Pipelines, Vector Embeddings (sentence-transformers), Google Gemini API, NumPy
- **Frontend:** React, TailwindCSS
- **Databases:** MongoDB, MongoDB Atlas, MySQL
- **Tools & Platforms:** Git, GitHub, Postman, Node-RED, AWS (EC2, S3 – basic), Linux CLI
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, System Design, IoT, Agile/Scrum

Experience

JavaScript Intern | *Atyantik Technologies*

May 2025 – Jul 2025

- Integrated RESTful APIs into client-facing features, reducing average data fetch latency by **20%** and improving overall user experience across the application.
- Engineered reusable React component library and optimized Redux state management, improving codebase scalability and reducing redundant code by approximately 30%.
- Collaborated in two-week Agile sprints with senior developers; maintained clean commit history through Git branching, pull requests, and structured code reviews on GitHub.

Projects

Study Copilot – RAG Backend

[🔗 Source Code](#)

- Built a Retrieval-Augmented Generation (RAG) backend in Python that ingests study documents, generates vector embeddings using **sentence-transformers** (all-MiniLM-L6-v2), and retrieves semantically relevant chunks at query time via cosine similarity search.
- Engineered a structured prompt pipeline that passes retrieved context to the Google Gemini API with source-citation instructions, ensuring factually grounded, hallucination-resistant answers.
- Designed a modular architecture across **embeddings.py**, **vector_store.py**, **rag_pipeline.py**, and **ingest.py** with clean separation of concerns; validated end-to-end with a dedicated **tester.py** suite.
- **Tech Stack:** Python, RAG, sentence-transformers, Google Gemini API, NumPy, Vector Search

HotReload – Go CLI Hot-Reloading Tool

[🔗 Source Code](#)

- Built a cross-platform CLI tool in Go that watches source files for changes, automatically rebuilds the binary, and restarts the server process — eliminating manual restart overhead during development.
- Architected a modular package structure (**internal/watcher**, **internal/builder**, **internal/process**) and used Go's **os/exec** and **syscall** for non-blocking process management with graceful termination.
- Validated the full reload lifecycle using a bundled **testserver** HTTP demo, enabling reproducible testing without external dependencies.
- **Tech Stack:** Go, CLI, File System Watching, Process Management, **os/exec**

Awards & Leadership

- **Best Director** – Riviera 2025 Inter-College Stage-Play Competition, VIT Chennai; led and coordinated a 15+ member cast and crew from pre-production through live performance. *Feb 2025*
- **All-Rounder Excellence Award** – St. Patrick's Higher Secondary School, Asansol *May 2023*